

QuanHack Technology Walkthrough

A detailed PDF guide to the AgentSDK-centered, AI-native architecture built for the separate technology prize. The focus is technical merit: agent orchestration, read-only tools, Anthropic critique, guardrails, traceability, and demo reproducibility.

SUBMISSION
PASS

RUBRIC
100/100

RED TEAM
PASS

REHEARSAL
PASS

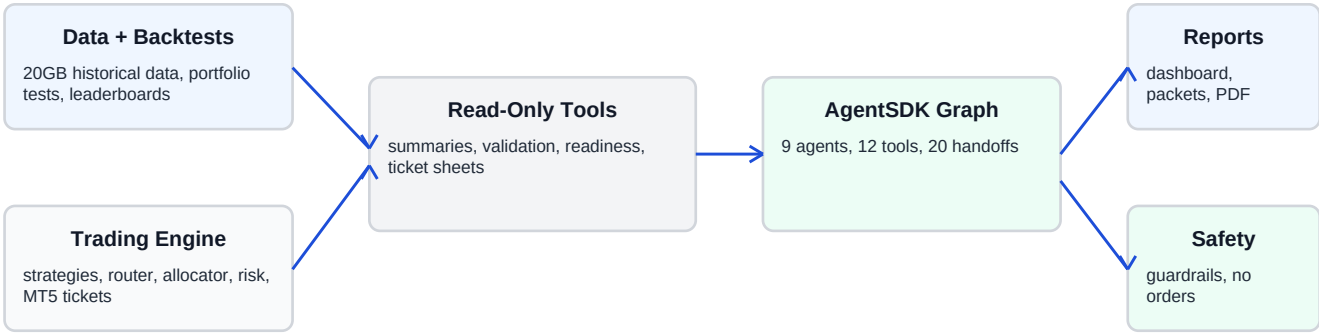


Figure 1. Technology stack overview: trading research remains deterministic; the AgentSDK layer governs evidence, critique, safety, and presentation.

- Built for the separate technology prize, not only trading P&L.
- Default demo path is offline and credit-safe.
- Agent tools are read-only; no MT5 order placement is exposed to models.
- Final bundle generated: 2026-06-27T10:13:42.763095+00:00

1. Executive Snapshot

QuanHack has been wrapped as a technology-prize submission around a safe agentic control plane. The trading engine still supplies research artifacts, backtests, risk outputs, dashboards, and MT5 ticket sheets. The new technology layer makes those artifacts inspectable by specialist agents and proves the boundary between AI reasoning and trading authority.

Component	Status	Meaning
Submission bundle	PASS	Final artifact manifest and command index.
Rubric	PASS	Score 100/100 across technology-prize axes.
Red-team report	PASS	8 skeptical judge challenges.
Demo rehearsal	PASS	8 offline demo steps.
Guardrails	PASS	8 executable safety checks.
Topology	PASS	9 agents and 12 tools.
Trace replay	PASS	36 span-like events.

What the judges should understand in one minute

- The system is no longer just a collection of scripts. It has a multi-agent control plane with explicit roles, handoffs, tools, and guardrails.
- OpenAI AgentSDK is the orchestration center: agents inspect evidence through typed, read-only function tools.
- Anthropic credits have a defined role: an independent critic reviews risk, overfitting, and presentation quality without trading authority.
- The safety model is executable. The project can prove that AI tools cannot place MT5 orders and cannot silently spend model credits.
- The demo is reproducible: `tech-prize-rehearse` and `tech-prize-submit` regenerate the proof pack offline.

2. System Architecture

The project is organized as a layered system. Lower layers do deterministic trading work: ingest data, run strategies, route signals through risk, backtest portfolios, and prepare MT5 ticket artifacts. The technology-prize layer sits above that engine. It does not replace the trading logic; it supervises and explains it.

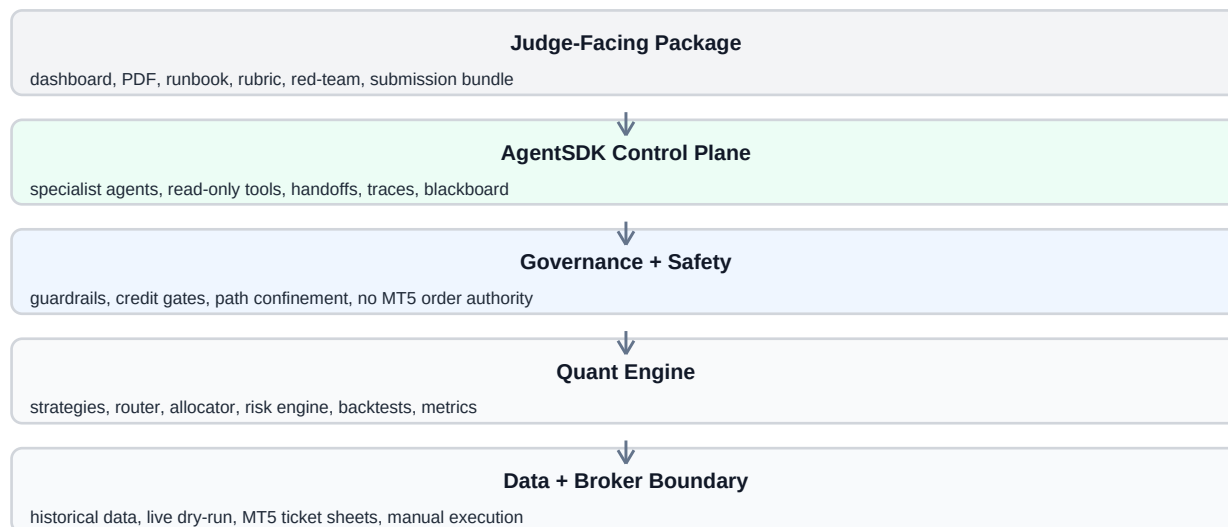


Figure 2. Layered architecture: deterministic quant engine below, agentic evidence and governance layer above.

Why this matters

A judge can inspect how the system is built without trusting a live model response. Every layer produces artifacts, and every technology-prize claim points back to files, hashes, reports, or executable checks.

3. AgentSDK Control Plane

The control plane defines 9 specialist agents, 12 read-only tool specs, and 20 validated handoffs. The Chief Trading Agent coordinates specialist roles instead of letting one prompt do everything.

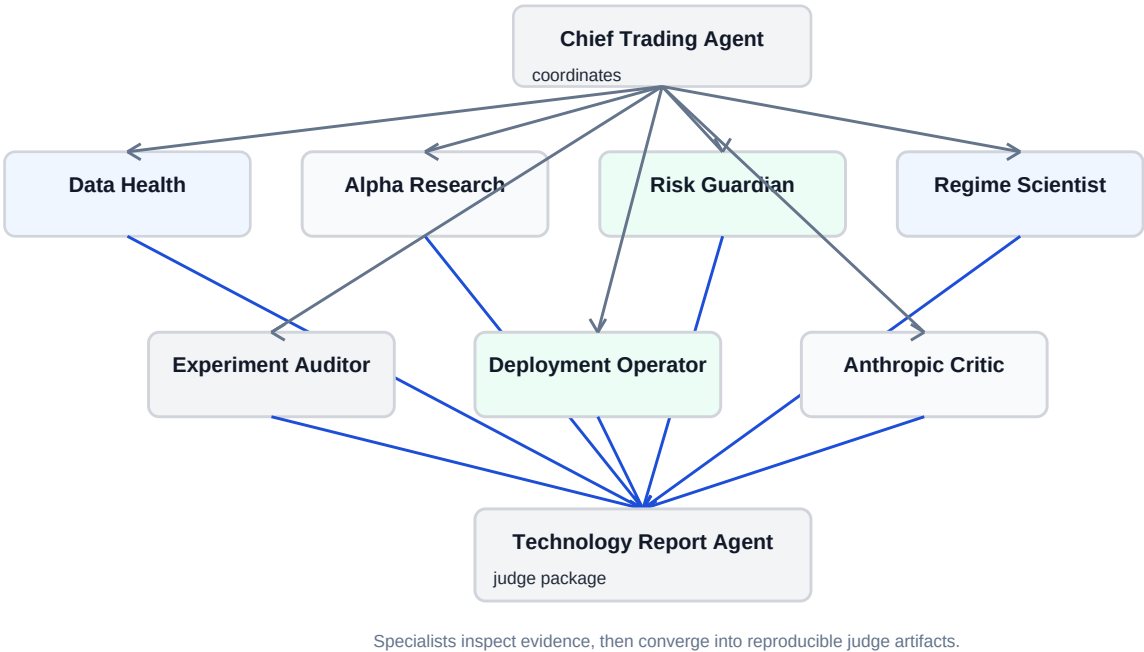


Figure 3. Agent graph: the Chief Trading Agent delegates to specialist roles and all roads lead to judge-facing reporting.

Agent	Technology role
Chief Trading Agent	Coordinates the evidence review and delegates to specialists.
Data Health Agent	Checks downloaded data, quote coverage, and market-data quality.
Alpha Research Agent	Reads research leaderboards, strategy comparisons, and walk-forward evidence.
Risk Guardian Agent	Inspects risk limits, guardrail status, MT5 boundary, and deployment safety.
Regime Scientist Agent	Explains regime-aware selector and handoff evidence.
Experiment Auditor Agent	Checks hashes, provenance, commands, and reproducibility.
Deployment Operator Agent	Summarizes dashboards, MT5 ticket sheets, and live dry-run readiness.
Anthropic Critic Agent	Independent critique of overfitting, risk, and narrative quality.
Technology Report Agent	Packages proof for judges.

4. Read-Only Function Tool Surface

The AgentSDK bridge exposes evidence tools, not trading controls. This is the main safety and technical-design point: models can inspect project artifacts, but they cannot route orders.

Tool	Purpose
summarize_research_artifacts	Summarize research/deployment files that support claims.
summarize_csv	Inspect bounded rows and schema from project-local CSV files.
validate_market_data_summary	Check price/quote data health.
summarize_experiment_leaderboard	Read walk-forward and candidate leaderboards.
build_hackathon_readiness_snapshot	Summarize competition readiness evidence.
summarize_mt5_ticket_sheet	Inspect manual MT5 ticket artifacts without order authority.
summarize_operator_dashboard_sources	Check live operator/dashboard source readiness.
build_technology_prize_judge_packet	Generate requirement-level evidence matrix.
run_agent_guardrail_suite	Run AI/broker safety checks.
analyze_agent_topology	Validate graph, handoffs, tools, and guardrails.
build_judge_demo_runbook	Build timed demo script and risk answers.
replay_agent_trace	Export local workflow as span-like trace data.

- All tools are read-only.
- Paths are confined to the project root.
- No MT5 order-placement tool is registered.
- Online model calls are disabled unless explicitly armed.

5. AI-Native Workflow And Traceability

The project treats agent reasoning as inspectable state. The local workflow writes blackboard entries, records handoffs, and exports an offline trace replay. That makes the architecture demonstrable even without spending API credits.

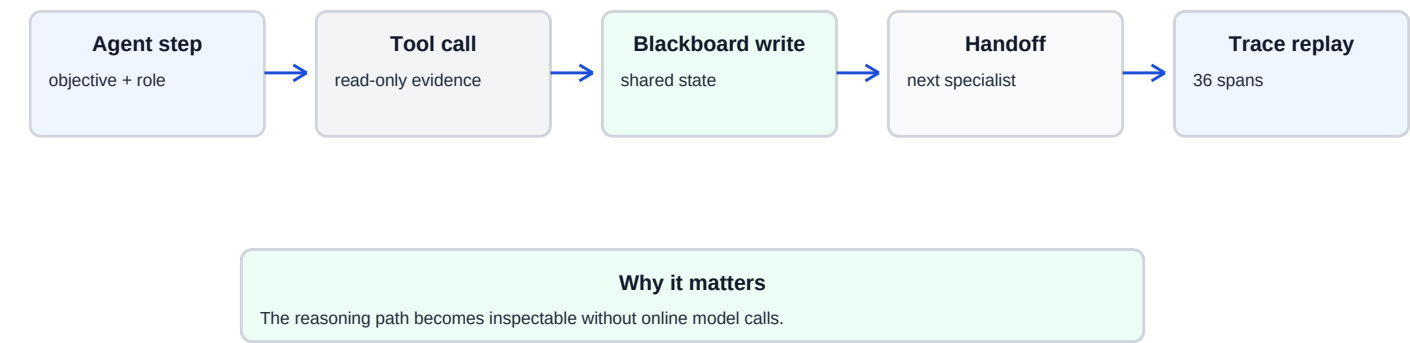


Figure 4. The workflow produces agent steps, tool calls, blackboard writes, and handoffs, then exports trace-like spans.

Trace kind	Count
agent_step	9
blackboard_write	9
handoff	8
tool_call	9
trace_root	1

The most important detail is not the count itself. It is that a judge can see how specialist agents move from evidence to claims through explicit state transitions.

6. Safety, MT5 Boundary, And Credit Controls

The technology prize judges may ask whether an AI agent can accidentally trade or spend credits. The answer is no by default. The project makes both boundaries executable: no broker authority is exposed to agents, and online OpenAI/Anthropic calls require explicit flags.

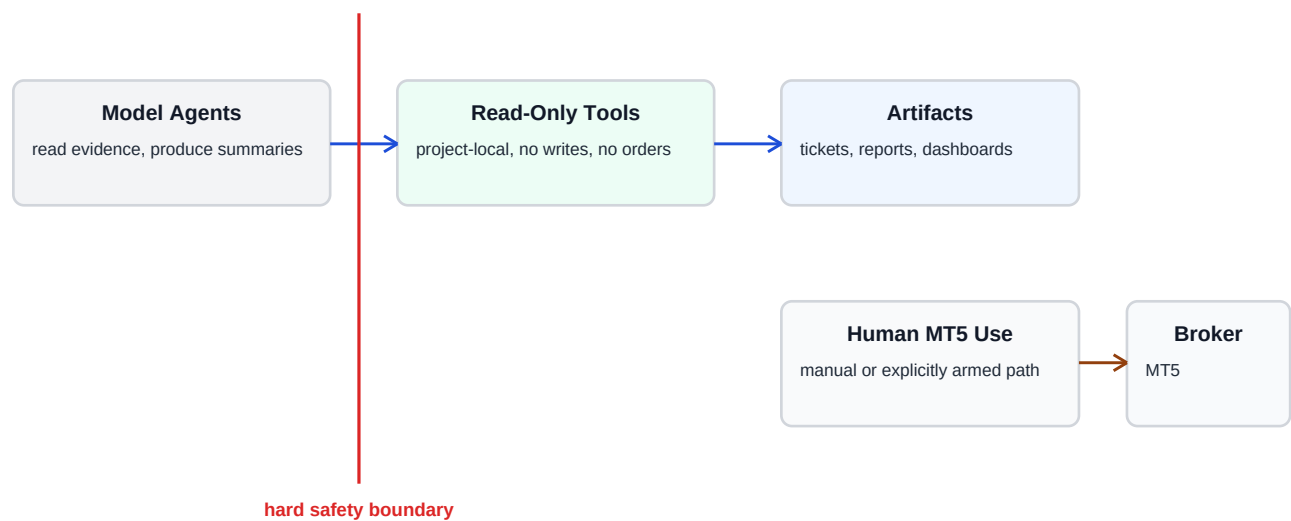


Figure 5. Safety boundary: models inspect artifacts; humans or separately armed systems control live MT5 execution.

Guardrail	Status	Details
Read-only AgentSDK tools	PASS	12 tool specs inspected; every tool must be read-only.
No broker order authority	PASS	Workflow blackboard order authority: none exposed to agents
Project-local path confinement	PASS	SDK CSV/artifact tools must reject paths outside the project root.
Online OpenAI calls require arming	PASS	AgentSDK Runner calls default to skipped/blocked unless explicitly allowed.
Online Anthropic calls require arming	PASS	Anthropic critic calls default to skipped/blocked unless explicitly allowed.
Anthropic remains critique-only	PASS	Anthropic Critic Agent must not approve trades or receive broker credentials.
Evidence before claims	PASS	7/7 required research/deployment artifacts are present.
Deterministic workflow trace	PASS	9 steps, 9 blackboard writes, 8 handoffs.

7. Anthropic Critic Layer

The Additional Anthropic Credits are not decorative. They have a defined technical role: independent critique. The Anthropic Critic Agent reviews overfitting risk, risk controls, and demo narrative while remaining outside broker authority.



Figure 6. Anthropic is used as a reviewer, not a trader. Its output improves quality assurance and judge readiness.

Design rules

- Anthropic online calls default to skipped unless `--allow-online-anthropic` is passed.
- The critic cannot approve trades and has no MT5 credentials.
- The critic reviews evidence produced by the deterministic system and AgentSDK control plane.
- This keeps multi-provider AI useful while preserving operational safety.

8. Evidence Evaluators

The final system includes evaluators that test the technology story from several angles. This is what makes the submission more than an architecture diagram: the claims are machine-checkable.

Evaluator	Status	What it proves
Topology	PASS	Tools are used, handoffs target real agents, and every tool is read-only.
Guardrails	PASS	No broker authority, path confinement, model-spend gates, and critic boundaries.
Trace replay	PASS	Agent steps, tool calls, blackboard writes, and handoffs are inspectable.
Rubric	PASS	100/100 score across technology-prize axes.
Red team	PASS	Skeptical judge objections are answered with evidence checks.
Rehearsal	PASS	Safe demo flow regenerates expected artifacts offline.

Red-team challenges

Challenge	Status	Skeptical question
No Hidden Trading Authority	PASS	Can the AI agent secretly place MT5 trades or bypass the risk engine?
Real AgentSDK Architecture	PASS	Is AgentSDK actually central here, or is this just a normal Python project with an agent label?
Anthropic Credits Have A Real Role	PASS	Where would Anthropic credits actually improve the system?
AI-Native, Not Script-Native	PASS	What makes this AI-native instead of many scripts wrapped in a dashboard?
No Surprise Model Spend	PASS	Could a demo command accidentally spend OpenAI or Anthropic credits?
Evidence And Provenance Survive Scrutiny	PASS	Can the judges verify claims from files instead of trusting the presentation?
Demo Can Be Reproduced Under Time Pressure	PASS	If the live demo gets rushed, is there a reliable proof path?
Technology Prize Story Is Separate From PnL	PASS	If trading P&L is weak, why is this still competitive for the separate technology prize?

9. Judge Demo Flow

The demo flow is intentionally offline by default. The goal is to avoid surprises during judging while still proving advanced technical architecture.

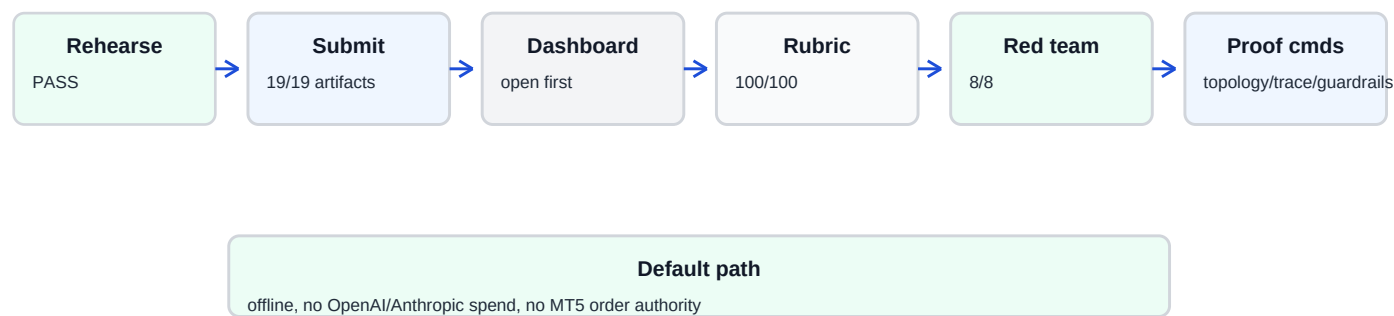


Figure 7. Recommended live flow: rehearse, submit, open dashboard, then run targeted proof commands on request.

Step	Command	Purpose
1	quanthack tech-prize-pack && quanthack tech-prize-dashboard	Regenerate the main scorecard and browser-first dashboard.
2	quanthack tech-prize-runbook	Regenerate the timed judge talk track.
3	quanthack tech-prize-topology	Validate the AgentSDK graph, handoffs, and tool authority.
4	quanthack tech-prize-trace	Export agent steps, tool calls, blackboard writes, and handoffs as spans.
5	quanthack tech-prize-guardrails	Verify read-only tools, path confinement, credit gates, and MT5 no-order authority.
6	quanthack tech-prize-judge-packet	Verify the prize requirements against source and report evidence.
7	quanthack tech-prize-rubric	Score the project against the technology-prize judging axes.
8	quanthack tech-prize-red-team	Answer skeptical judge questions with executable evidence checks.

10. Final Artifact Map

The final bundle is the handoff package. It tells the judge what to open first, which commands are safe, and which reports prove each claim.

Artifact	Status	Path
Open first: dashboard	OK	outputs/reports/technology_prize_dashboard.html
Judge demo runbook	OK	outputs/reports/technology_prize_demo_runbook.md
Judge demo rehearsal	OK	outputs/reports/technology_prize_demo_rehearsal.md
Judge demo rehearsal JSON	OK	outputs/reports/technology_prize_demo_rehearsal.json
Judge packet	OK	outputs/reports/technology_prize_judge_packet.md
Judge packet JSON	OK	outputs/reports/technology_prize_judge_packet.json
Technology prize rubric	OK	outputs/reports/technology_prize_rubric.md
Technology prize rubric JSON	OK	outputs/reports/technology_prize_rubric.json
Technology prize red team	OK	outputs/reports/technology_prize_red_team.md
Technology prize red team JSON	OK	outputs/reports/technology_prize_red_team.json
Demo pack	OK	outputs/reports/technology_prize_demo_pack.md
Demo pack JSON	OK	outputs/reports/technology_prize_demo_pack.json
AgentSDK topology	OK	outputs/reports/technology_prize_topology.md
AgentSDK trace replay	OK	outputs/reports/technology_prize_trace_replay.md
Agent guardrails	OK	outputs/reports/technology_prize_guardrails.md
Local workflow	OK	outputs/reports/technology_prize_workflow.md
Agent architecture report	OK	outputs/reports/technology_prize_agent_report.md
AgentSDK runner report	OK	outputs/reports/technology_prize_sdk_runner.md
Anthropic critic report	OK	outputs/reports/technology_prize_anthropic_critic.md

11. Appendix: Commands And Talking Points

Core commands

Command	Use
quanthack tech-prize-rehearse	Verifies the safe offline demo flow.
quanthack tech-prize-submit	Regenerates the final submission bundle and manifest.
open outputs/reports/technology_prize_dashboard.html	Opens the first judge-facing visual artifact.
quanthack tech-prize-rubric	Shows the scored prize framing.
quanthack tech-prize-red-team	Answers skeptical judge questions with evidence.
quanthack tech-prize-topology	Validates AgentSDK graph integrity.
quanthack tech-prize-trace	Shows trace-like agent spans.
quanthack tech-prize-guardrails	Shows executable AI/broker safety checks.

Closing pitch

- **QuanHack is not just a trading bot. It is an AI-native research and deployment control plane.**
- **AgentSDK coordinates specialist agents over real project evidence.**
- **Anthropic credits support independent critique rather than uncontrolled action.**
- **Guardrails prove that models cannot place trades or spend credits without explicit arming.**
- **The final demo is reproducible offline and backed by hashed artifacts.**